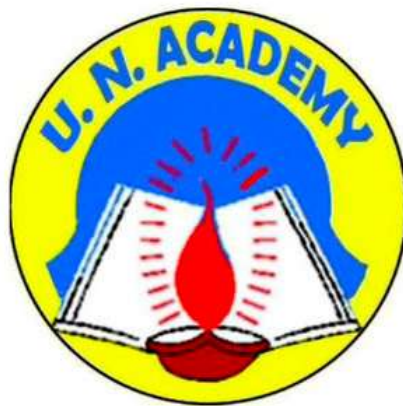


LONG-TERM TRENDS IN LAND SURFACE TEMPERATURE OF BONGAIGAON USING MULTI-TEMPORAL SATELLITE DATA

A project report submitted for the partial fulfillments for the

DEGREE OF BECHLOR OF ARTS IN GEOGRAPHY



Under the guidance of:
(Aruna Narzary)
Assistant Professor
Department of Geography

Submitted By:
Ansula Basumatary
B.A. 6th Semester
U.N. Academy, Kokrajhar
Affiliated to Bodoland University
Kokrajhar, 783370, BTR, Assam

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CHAPTER-1

1.1 Introduction:

Land Surface Temperature (LST) refers to the thermal energy emitted by the land surface, excluding any influence from the atmosphere. It is the key variable of the Earth's Surface as measured from either satellite based remote sensing instruments and ground-based sensors and are the crucial parameter in various fields such as climate studies, urban planning, agriculture and environmental monitoring, providing insights into local climate dynamics, land use changes and anthropogenic influences on the environment. LST is closely linked with global temperature (GT), sea surface temperature (SST) and land surface sea temperature (LSAT).

Remote sensing technology plays a crucial role in estimating land surface temperatures (LST). Because using remote sensing method has include various advantages like large area coverage, ability to estimate accurate temperature in accessible areas, very efficient and effective method, and lastly, is cost effective. Remote sensing using thermal scanning of various surface types, such as plants and soils, simultaneously reveals their temperature reactions to atmospheric and radiant influences. Each feature on the Earth's surface emits thermal radiation at different wavelengths based on its emissivity (E). Emissivity is the ratio of the spectral radiant emittance of a grey body to that of a blackbody at the same temperature. Accurate emissivity values are crucial for reliable comparisons among different land covers when retrieving Land Surface Temperature (LST) from thermal infrared (TIR) data. These variables provide insights into various Earth surface processes, surface-atmosphere interactions, and evapotranspiration. Estimating LST across the Earth's surface is essential for understanding the interactions between biotic and abiotic environmental components. LST directly influences crop productivity and impacts groundwater quality and temperature. This study aims to assess the performance of LST retrieval methods using different Land Surface Emissivity (LSE) models. To facilitate this work, several software were developed like QGIS, ArcGIS, Earth Engine etc where all the methods and models are analysed and provided as a resource for LST retrieval from Landsat and MODIS data.

Remote sensing techniques have revolutionized the field of Earth observation and monitoring by providing valuable information about various aspects of our Earth surface. In recent years, the use of remote sensing for monitoring and forecasting Land Surface Temperature (LST) has gained significant attention due to its importance in understanding climate change, urban heat island effects and agricultural activities with the increasing availability of satellite data, remote sensing techniques have emerged as powerful tools for monitoring and forecasting LST on both regional and global scales.

1.2. Significance of the Study:

Studying the land surface temperature (LST) of Bongaigaon is pivotal for understanding and addressing a multitude of socio-environmental challenges. Bongaigaon's rapid urbanization and industrialization make it a focal point for assessing the impacts of human activities on local climate dynamics. Studying the land surface temperature of Bongaigaon is really important for a few reasons. It helps us understand how hot or cold the land there is, which can tell us a lot about the local climate. This is super useful for farmers, urban planners, and anyone who needs to know about the weather in that area. And looking at land surface temperature can help us figure out if there are any changes happening in the environment. For example, if the land is getting hotter over time, it might mean that there's something going on with the climate or maybe there's more development happening in the area. And also studying the land surface temperature of Bongaigaon can also help us understand how cities and urban areas affect the temperature of the land. This is called the urban heat island effect. Basically, cities tend to be hotter than rural areas because of all the buildings and roads that absorb heat during the day and release it at night. Overall, studying the land surface temperature of Bongaigaon is important because it helps us understand the local climate, track changes in the environment, and learn more about how cities affect temperature.

1.3. Objectives:

- a) Comparative analysis of LST of Bongaigaon through remote sensing techniques.
- b) Forecasting LST of the Bongaigaon

1.4. Statement of the Problem:

The Bongaigaon district of Assam faces a complex and multifaceted set of challenges stemming from the significant variation in its land surface temperature. As the largest town and an important commercial and administrative centre in the district, Bongaigaon is particularly susceptible to the urban heat island effect, a phenomenon where the conversion of natural land cover to urban and built-up areas lead to evaluated temperature within the city compared to the surrounding rural regions. This urban warming can have far reaching implications, affecting the overall human comfort and well-being of local population, increasing energy demands for cooling during the hot summer months, and altering the local microclimate in ways that can disrupt the delicate balance of the ecosystem. The impacts of these temperature variations, however, it extends well beyond the urban centre of Bongaigaon, as the district is also an important agricultural region, known for the production of crop such as rice, jute and mustard. Fluctuations in land surface temperature can disrupt traditional growing seasons, reduce crop yields, and alter the overall sustainability of certain agricultural areas, potentially jeopardising food production and food security for the local population. Bongaigaon is home to several ecologically significant wetlands including the renowned Manas National Park, a UNESCO world Heritage site.

Changes in land surface temperature can profoundly affect the hydrology and

ecology of these wetland ecosystems, leading to habitat degradation, loss of biodiversity and adverse impacts on the delicate balance of the region's natural resources. The availability of and quality of water resources, which are essential for bothersome agricultural and domestic use, may also be impacted by alterations in the water cycle driven by temperature variations. This multifaceted and interconnected problems have far-reaching consequences for the district, affecting various aspects of local economy, environment, and the well-being of the population. Addressing these issues will require a comprehensive, collaborative approach involving local authorities, research institutions, and community stakeholders to develop targeted policies, implemented climate resilient interventions and enhance the district's overall resilience to complex and far-reaching impacts of land surface temperature changes. This will involve a range of strategies, such as urban planning to mitigate the urban heat island effects, sustainable agricultural practices to adapt to changing climate conditions ecosystems based on management of wetlands and natural resources and the environment of vulnerable communities to build their adaptive capability and resilience

1.5. REVIEW OF THE RELEVANT LITERATURES:

The topic has very vast literatures some of them are mentioned below:

Markus Neteler (2010) in his journal of LST "Estimating Daily Land Surface Temperatures in Mountainous Environments by Reconstructed MODIS LST DATA" stated that the availability of new satellites data with high temporal resolution offers fresh opportunities for remote sensing in ecology, epidemiology and other fields. He also described that the new reconstructed LST time series are reducing the gap between high spatial resolution and high temporal resolution.

A.Benali, A.C. Carvalho, J.P. Nunes, N. Carvalhais and A. Santos (2012) in their journal of Air Surface Temperature "Estimating air surface temperature in Portugal using MODIS LST DATA" they showed that the weekly average maximum, minimum, average air temperature can be accurately estimated using remote sensing techniques. They stated that the developed dataset accurately both intra-inter annual and spatial pattern of air temperature.

CHAPTER-2: SELECTION OF THE STUDY AREA:

2.1. Location of the Study Area:

Bongaigaon district is situated in the northeastern part of India and district of Assam. It is located in 26°26' N latitudes and 90°31'E longitudes. Bongaigaon district occupies an area of 1,093 square kilometres (422 sq. mi). It is bordered by Barpeta to the east, the Brahmaputra River to the south, and Kokrajhar to the north and west. Additionally, it shares an international border with Bhutan to the north.

2.2. Physiography:

The physiography of Bongaigaon district varies from plains to hilly terrains. While the southern part is predominantly flat due to its proximity to Brahmaputra River, the northern and western part is gradually ascends into hilly terrain, blending into the foothills of the Himalayas. The district is characterised by the fertile agricultural plains, interspersed with patches of forested areas.

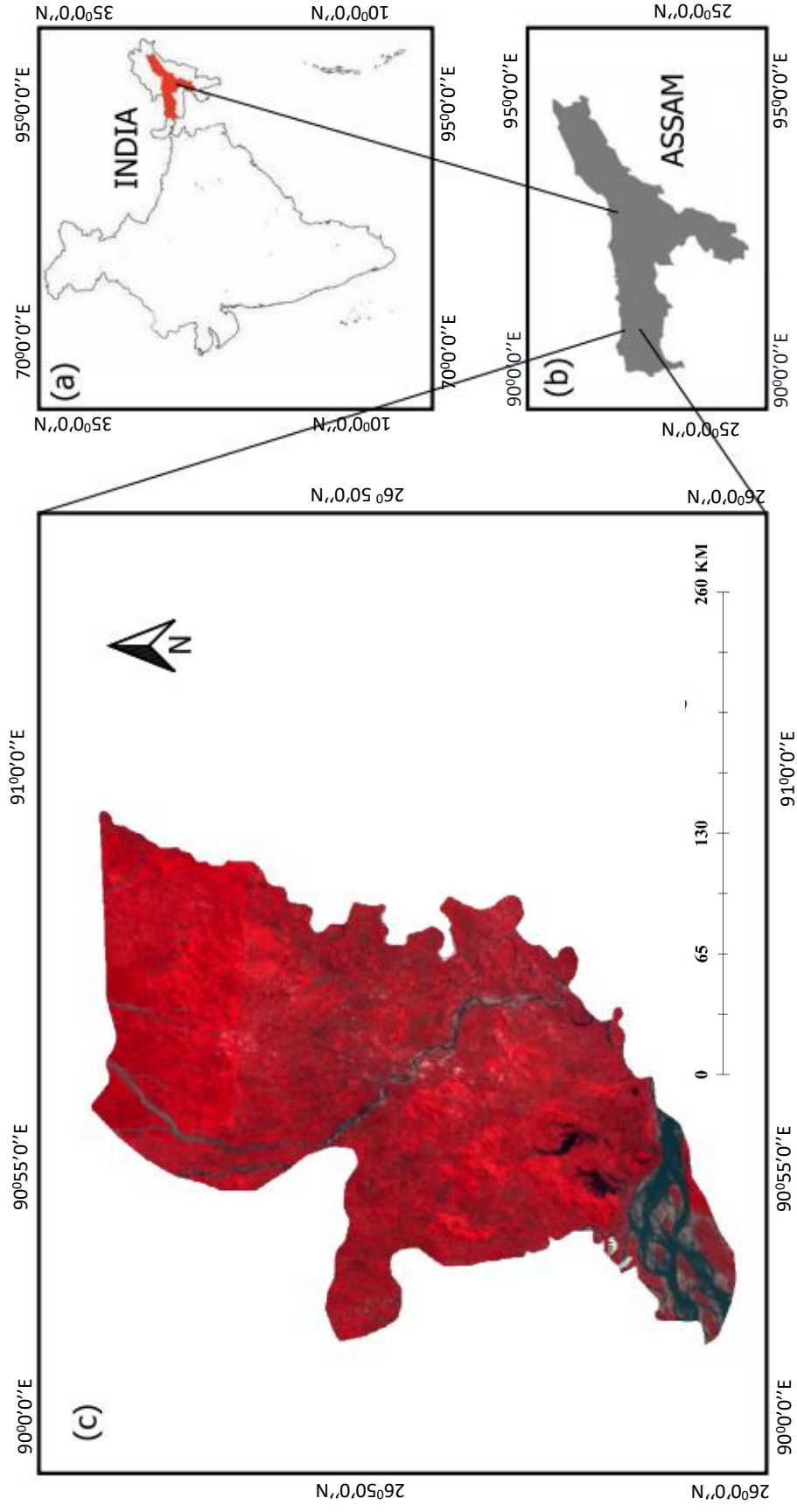
2.3. Climate:

Bongaigaon district experiences a subtropical climate characterised by hot and humid summer and relatively mild winters. The monsoon season typically spans from June to September, bringing heavy rainfall, which is vital for agriculture in the region. Summers are hot with temperature often soaring, while winters are relatively cooler but still comfortable. The climate supports a diverse range of flora and fauna, contributing to the district's ecological richness.

2.4. Vegetation:

The vegetation in Bongaigaon district of Assam showcases a diverse range of flora, owing to its varied physiography and climatic conditions. The district's plains are dominated by lush paddy fields, interspersed with wetlands and marshy areas, which support a rich variety of aquatic plants. Along the riverbanks of the Brahmaputra, you can find extensive stretches of alluvial grasslands and riverine forests, providing habitat to numerous species of birds, mammals, and reptiles. As the terrain gradually ascends towards the northern and western parts of the district, the vegetation transitions into subtropical forests, characterized by a mix of deciduous and evergreen trees. Sal forests are prevalent in some areas, while bamboo groves are common in the hilly terrain. The district's forests are home to a diverse array of flora, including valuable timber species like teak, sal, and mahogany, alongside a variety of medicinal plants and herbs. Additionally, the region supports a rich diversity of orchids, ferns, and other epiphytic plants.

**REPRESENTING THE LOCATION OF THE STUDY AREA, BONGAIGAON DISTRICT,
ASSAM, INDIA
IN FALSE COLOUR COMPOSITE (FCC)
LANDSAT 8 OLI/TIRS**



2.5. Drainage System:

The drainage system of Bongaigaon district is primarily influenced by the Brahmaputra River and its tributaries, along with other smaller rivers and streams. Here's an overview:

- **Brahmaputra River:** The Brahmaputra River, one of the major rivers in Asia, flows along the southern boundary of Bongaigaon district. It serves as a lifeline for the region, providing water for irrigation, transportation, and other socio-economic activities. The river originates in the Tibetan plateau and enters Assam through the eastern part of the state. In Bongaigaon, the Brahmaputra flows in a southward direction, forming a natural boundary between the district and neighboring Barpeta district. Its extensive network of tributaries and distributaries contributes to the overall drainage pattern of the region.
- **Tributaries:** Several tributaries of the Brahmaputra River traverse through Bongaigaon district, draining water from the surrounding hills and plains. Some of the major tributaries include the Aie River, Champabati River, and Beki River. These tributaries play a vital role in replenishing the water resources of the region and supporting agriculture and other livelihood activities.
- **Other Rivers and Streams:** Apart from the Brahmaputra and its tributaries, Bongaigaon district is also traversed by other smaller rivers and streams. These water bodies contribute to the overall drainage network of the district, facilitating the discharge of rainwater and runoff into the Brahmaputra River and its tributaries.
- **Wetlands and Marshes:** The district is home to numerous wetlands, marshes, and oxbow lakes, which act as natural reservoirs and detention basins during periods of heavy rainfall and flooding. These wetlands play a crucial role in regulating the flow of water and maintaining hydrological balance in the region.

Overall, the drainage system of Bongaigaon district is characterized by a network of rivers, tributaries, wetlands, and marshes, which collectively contribute to the sustainable management of water resources and the ecological health of the region.

CHAPTER-3: MATERIAL AND METHODOLOGIES:

In this study, we analysed Landsat satellite imagery with a moderate resolution of 30 meters. The methodology carried out here using Google Earth Engine as well as QGIS. It was used to perform image clipping, estimation of LST, derivation of vegetation indices and for the compilations of maps. Additionally, we cross-referenced our results with high-resolution imagery from Google Earth to assess the accuracy of our map creating. The following Table 1 is the Landsat data that are took to create images or map

Table 1. Landsat 5and 8 data (sensors and resolution)

Date	Sensors	Path/Row	Resolution
2003	Landsat 5 TM	137/42	30
2013	Landsat 8 OLI/TIRS	137/42	30
2023	Landsat 8 OLI/TIRS	137/42	30

We also used Modis data to create Land Surface Temperature of the region of interest. Entire methodology is given in flow chart fig: 1

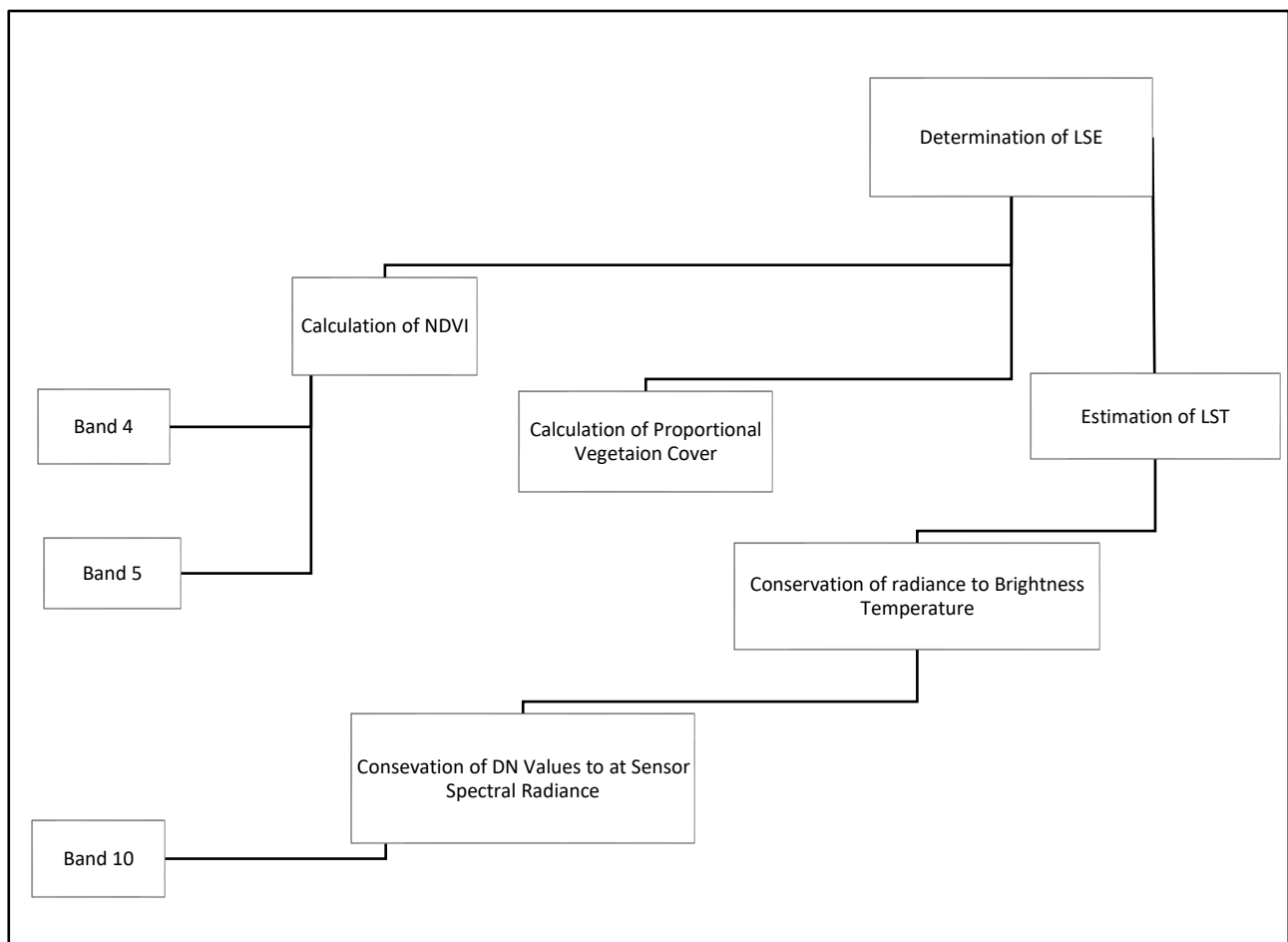


Fig 1: Flow diagram of LST retrieval

Land surface temperature is estimated using Landsat imagery from two different time periods. These images are obtained from the website (<https://developers.google.com/earth-engine/datasets>). The data provider has already applied radiometric corrections to the images. However, before calculating the surface temperature, the images need to be converted from digital number (DN), which is a dimensionless unit, to Top of Atmosphere (TOA) radiance. TOA radiance represents the actual reflectance from the Earth's surface. This conversion is essential because the DN does not correspond to any physical quantity. The subsequent steps involve using specific formulas to convert the TOA radiance into surface temperature.

3.1. LST Retrieval:

In order to achieve high-accuracy land surface temperature (LST) extraction, the split-window algorithm was employed. This process leveraged the LST_Day_1km band from MODIS (MOD11A1 V6.1) which provides daily land surface temperature (LST) and emissivity values in a 1200 x 1200 kilometer grid of the Earth's surface. The accuracy of these measurements can be compromised by atmospheric components, including water vapor, clouds or aerosols. Consequently, an atmospheric correction is necessary. This correction uses supplementary meteorological data—such as relative humidity and temperature profiles—to account for and remove the atmospheric influences from the thermal bands during LST calculation.

3.1.1. Top-of-atmosphere (TOA) reflectance calculation:

TOA reflectance refers to the fraction of sunlight that is reflected by the earth's surface and atmosphere back to the satellite. To estimate Land Surface Temperature (LST), it is crucial to understand surface reflectance across various spectral bands. For this purpose, Top-of-Atmosphere (TOA) reflectance is calculated using data from both Landsat 8 and MODIS. In the case of Landsat 8, raw digital data from the visible and near-infrared bands are converted into TOA reflectance values through atmospheric correction techniques, specifically using the Landsat Surface Reflectance Code (LaSRC). Similarly, MODIS data, including the LST_Day_1km band, undergo atmospheric correction to derive accurate surface reflectance values. These corrected reflectance values are essential for precise LST estimation. The following equation is used for converting the raw data (Qcal) in to spectral radiation ($L\lambda$):

$$L\lambda = ML \times Qcal + AL$$

Where:

- **$L\lambda$** is the spectral radiation in watts per square meter per steradian per meter ($W/m^2/sr/m$).
- **ML** is the multiplicative rescaling factor the specific spectral band.
- **Qcal** is the raw, quantized, and calibrated pixel values from the satellite's sensor.
- **AL** is the additive rescaling factor for the specific spectral band.

3.1.2. Top of Atmosphere (TOA) Brightness Temperature Conversion:

Brightness temperature is a measure of the thermal emission of an object as observed by a remote sensing instrument. The radiance measurements need to be converted into brightness temperature (T_b), which is the temperature of a blackbody (an idealized physical body that absorbs all incident electromagnetic radiation and re-emits it perfectly) would emit the same amount of radiative power per unit area at a specific wavelength or frequency as observed. In simpler terms, it is a way to quantify the amount of thermal energy being emitted by an object based on its radiance. The following equation is used for converting the Radiance in to the brightness temperature (T_b):

$$T_b = \frac{K2}{\ln(1 + \frac{K1}{L})}$$

Where:

- **T_b** is the brightness temperature in Kelvin (K).
- **$K1$** and **$K2$** are calibration constants specific to the thermal band being used.
- **L** is the measurement radiance value.

Brightness temperature provides a direct estimate of the surface temperature of the Earth. This is crucial for applications such as monitoring urban heat islands, studying climate change, and managing agricultural practices. It helps in understanding temperature variations over land and ocean surfaces, which is important for weather forecasting, climate models, and studying ocean currents.

3.1.3. Normalized Difference Vegetation Index (NDVI) Calculation:

The Normalized Difference Vegetation Index (NDVI) is a widely used remote sensing index that quantifies vegetation health and density based on how vegetation reflects light at certain wavelengths. To calculate the NDVI of the study area we used Landsat 8 OLI (Operational Land Imager) satellite image or data which is available on the Earth Engine Data Catalog and done in the Google Earth Engine. The NDVI formula is as follows:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

Where:

- **NIR** (Near-Infrared Light): Healthy vegetation reflects a large amount of near-infrared light.
- **Red Light**: Vegetation absorbs a significant portion of red light for photosynthesis

This formula calculates the NDVI value by comparing the reflectance of near-Infrared light to red light. Healthy vegetation reflects more near-infrared light and less red light, resulting in higher NDVI values, while non-vegetated surfaces reflect more red light and less near-infrared light, resulting in lower NDVI values. In general, NDVI uses satellite data to measure vegetation cover by analyzing the differences in reflected light in the red and near-infrared spectra.

The resulting NDVI value ranges from -1 to +1:

- Values close to +1 indicate high density and healthy vegetation.
- Values near 0 indicate barren areas of rock, sand, or urban regions.
- Values approaching -1 typically indicate water bodies.

3.1.4: Calculation of Proportion of Vegetation (Pv):

The Proportion of Vegetation (Pv) is a measure used in remote sensing to quantify the fraction of an area that is covered by vegetation. It is an important parameter in environmental studies and land surface analysis, including the estimation of Land Surface Temperature (LST). By accounting for the cooling effect of vegetation, Pv enhances the accuracy of LST retrieval. It does this by considering the fractional coverage of greenery within a given pixel or region. Pv is calculated as follows:

$$Pv = \{(NDVI - NDVI_{min}) / (NDVI_{max} - NDVI_{min})\}^2$$

squaring the normalized difference index makes it easier to differentiate between vegetated and non-vegetated areas and makes the index more sensitive to changes in vegetation. By calculating PV, essentially normalize the NDVI value and square it to get a value that represents the proportion of vegetation within the pixel. The result is a value between 0 and 1, where 0 means no vegetation and 1 means full vegetation.

3.1.5. Land Surface Emissivity (LSE):

Land Surface Emissivity (LSE) is a measure of the efficiency with which a surface emits thermal radiation compared to a blackbody at the same temperature. It is a crucial parameter in remote sensing and thermal infrared (TIR) applications because it influences the thermal radiation observed by sensors. LSE is crucial for accurately estimating Land Surface Temperature (LST) because LST is derived from the thermal infrared radiance measured by satellites. Since the radiance detected by a sensor depends on both the temperature of the surface and its emissivity, accurate LSE values are necessary to correct the measured radiance. Different surfaces (e.g., water, vegetation, urban areas) have different emissivities. These variations need to be accounted for to prevent errors in temperature estimation. Knowing the emissivity helps in identifying different land cover types, which is important for various environmental and climatic studies. The calculation of Land Surface Emissivity (LSE) involves understanding and using the Normalized Difference Vegetation Index (NDVI) values to estimate the proportion of vegetation (PV) on the Earth's surface. Here's a step-by-step explanation of the process and the formulas used:

$$E = 0.004 \times PV + 0.986$$

Where:

- **E:** This is the land surface emissivity, which quantifies how effectively the Earth's surface radiates heat.
- **Pv:** This is the proportion of vegetation calculated in the previous step.

- The formula shows that LSE increases slightly with an increase in vegetation proportion. The constants (0.004 and 0.986) are derived from empirical studies and indicate that even bare soil (PV = 0) has an emissivity close to 0.986.

This process helps in understanding the thermal properties of the earth's surface, which is crucial for various applications, including climate modeling, environmental monitoring, and agricultural assessments.

3.1.6. Land Surface Temperature:

Land Surface Temperature (LST) is a crucial parameter in remote sensing that measures the temperature of the Earth's surface. This measurement is important for various applications, including climate studies, agricultural monitoring, and urban planning. LST is derived from thermal infrared (TIR) radiation emitted by the Earth's surface. Remote sensing satellites equipped with TIR sensors detected this radiation. The amount of emitted radiation depends on the surface temperature and its emissivity, which is the efficiency with which a surface emits thermal radiation. Sensors like MODIS (Moderate Resolution Imaging Spectroradiometer) on NASA's Terra and Aqua satellites, Landsat's Thermal Infrared Sensor (TIRS), and Sentinel-3's SLSTR (Sea and Land Surface Temperature Radiometer) are commonly used for LST measurements. The MODIS LST retrieval algorithm uses a split-window approach, which involves two thermal infrared bands (usually 31 and 32). A common formula used for MODIS LST estimation is:

$$LST = a + b(T_{31} + T_{32}) + c(T_{31} - T_{32})^2 + d\frac{T_{31} - T_{32}}{\varepsilon} + e(1 - \varepsilon)$$

Where:

- T_{31} and T_{32} are the brightness temperatures from MODIS bands 31 and 32, respectively.
- ε is the average emissivity.
- a , b , c , d , and e are coefficients determined from regression analysis based on the sensor characteristics and atmospheric conditions.

In our case, we use Google Earth Engine to calculate or estimate the Land Surface Temperature (LST). We just put some code and get all the needed information or images.

CHAPTER-4: RESULTS AND DISCLUSION

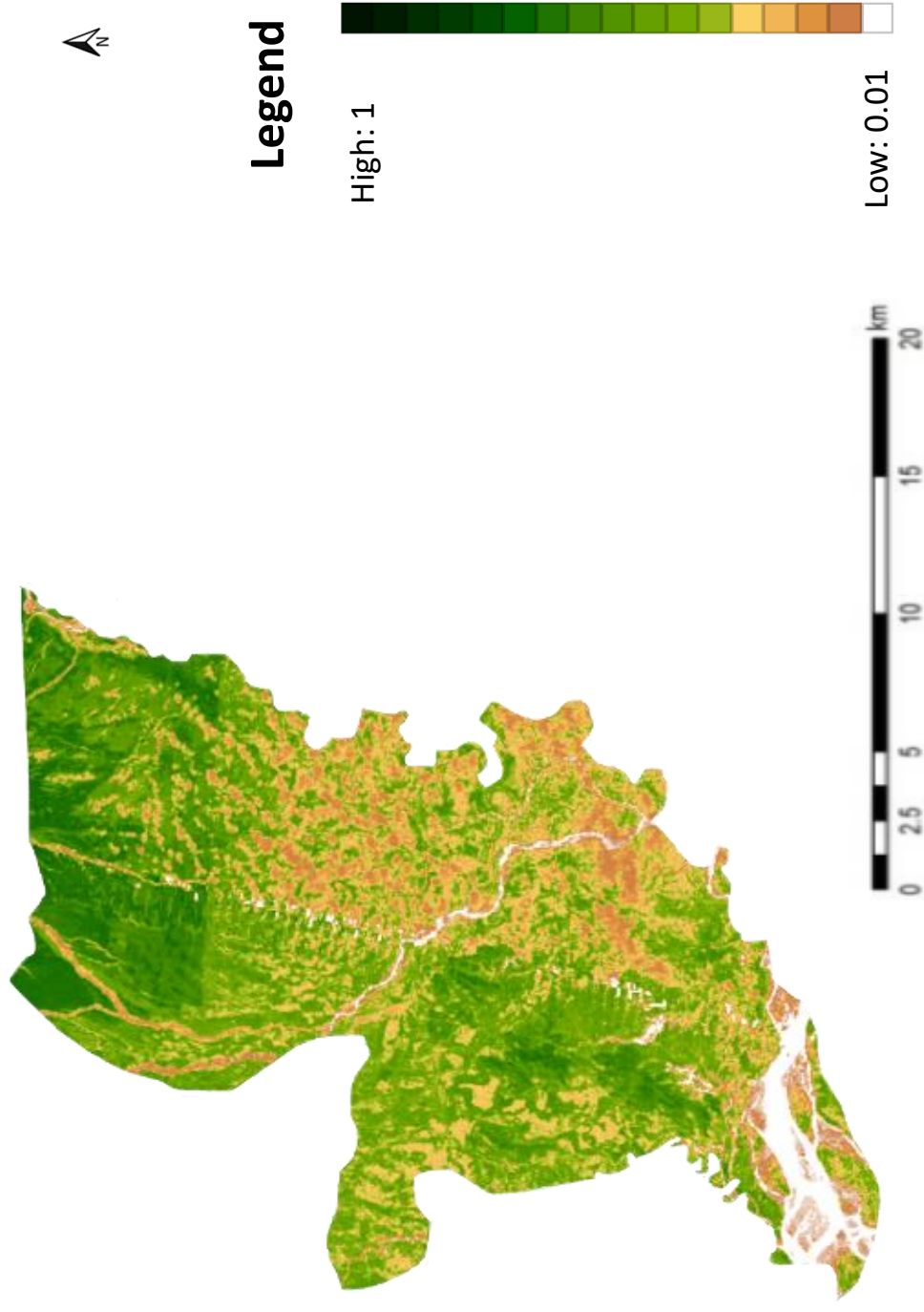
4.1. Analysis of Normalised Differences Vegetation Index (NDVI):

The NDVI (Normalized Difference Vegetation Index) map for the year 2003 indicates that the NDVI values range between -0.01 and 1. The map reveals that the central region extending to the southeastern part of the selected area exhibits low NDVI values, typically ranging from 0.1 to 0.25. This is primarily due to the presence of built-up areas or settlements, which are represented in red or yellow on the map. In contrast, the northern part of the district, which falls under the Himalayan foothills, shows significantly higher NDVI values compared to other regions, with values exceeding 0.6. Additionally, areas covered by water bodies display notably low NDVI values, ranging from -0.01 to 0, as illustrated on the 2003 NDVI map.

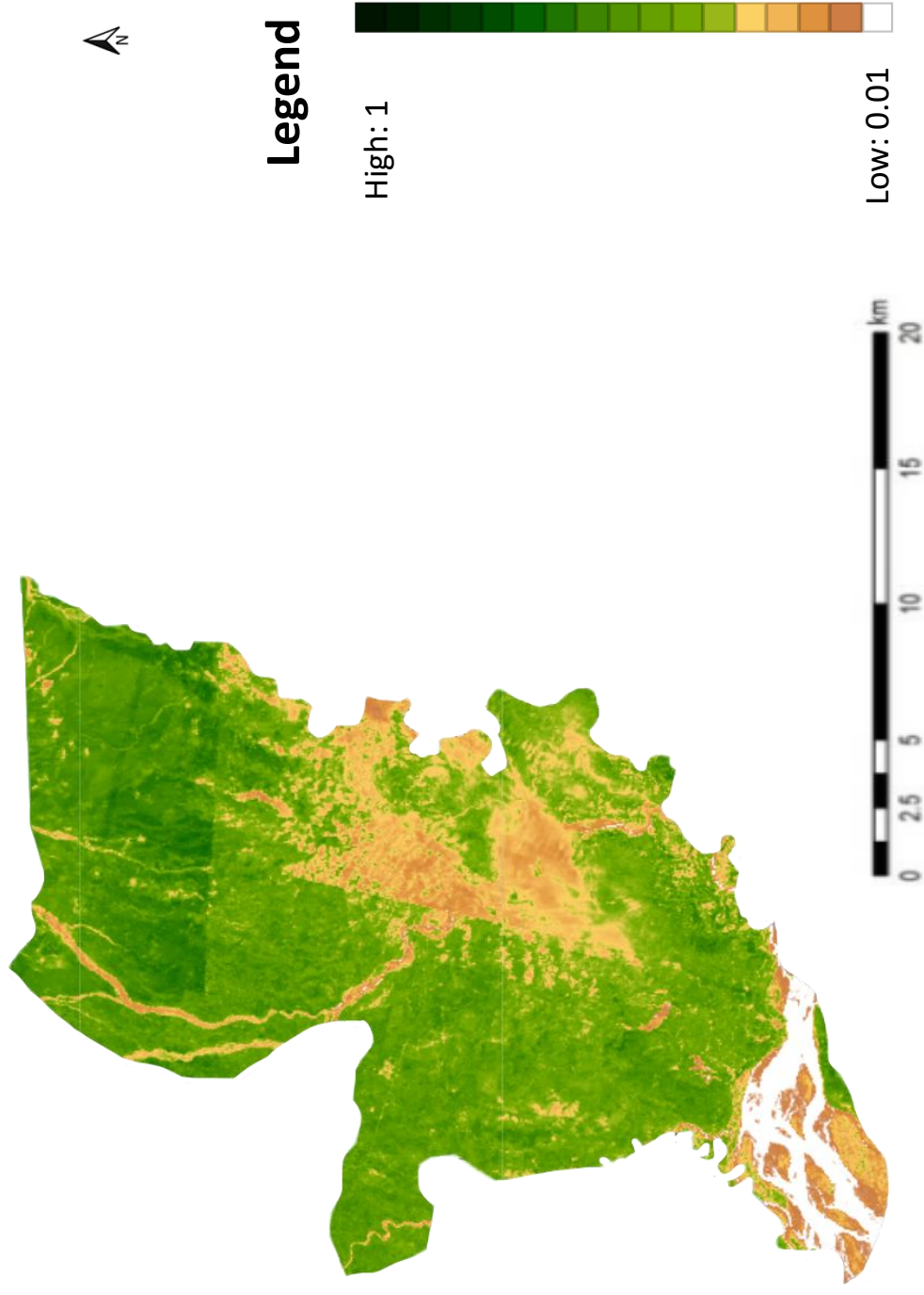
Moving forward to 2013, the NDVI map for this year also demonstrates values between -0.01 and 1. However, a distinct change is observed when comparing the data from 2013 to that of 2003. While the general characteristics remain the same, there is a noticeable shift in the NDVI values across certain parts of the region. Upon examining the 2013 map, it becomes evident that the extent of barren land or areas with low NDVI values has decreased. This suggests a slight increase in vegetation density, indicating an improvement in vegetative cover over the decade.

The NDVI map for 2023 continues to reflect a value range of -0.01 to 1. The resulting map displays moderate to high NDVI values across the region. A closer look at the map reveals that the trend observed in 2013 has persisted, with the vegetation density either maintaining or slightly improving. The areas with low NDVI values have further diminished, and the regions with higher vegetation density have become more prominent. This progression suggests a continuing enhancement in the vegetative cover and overall health of the region's ecosystem over the two-decade span from 2003 to 2023.

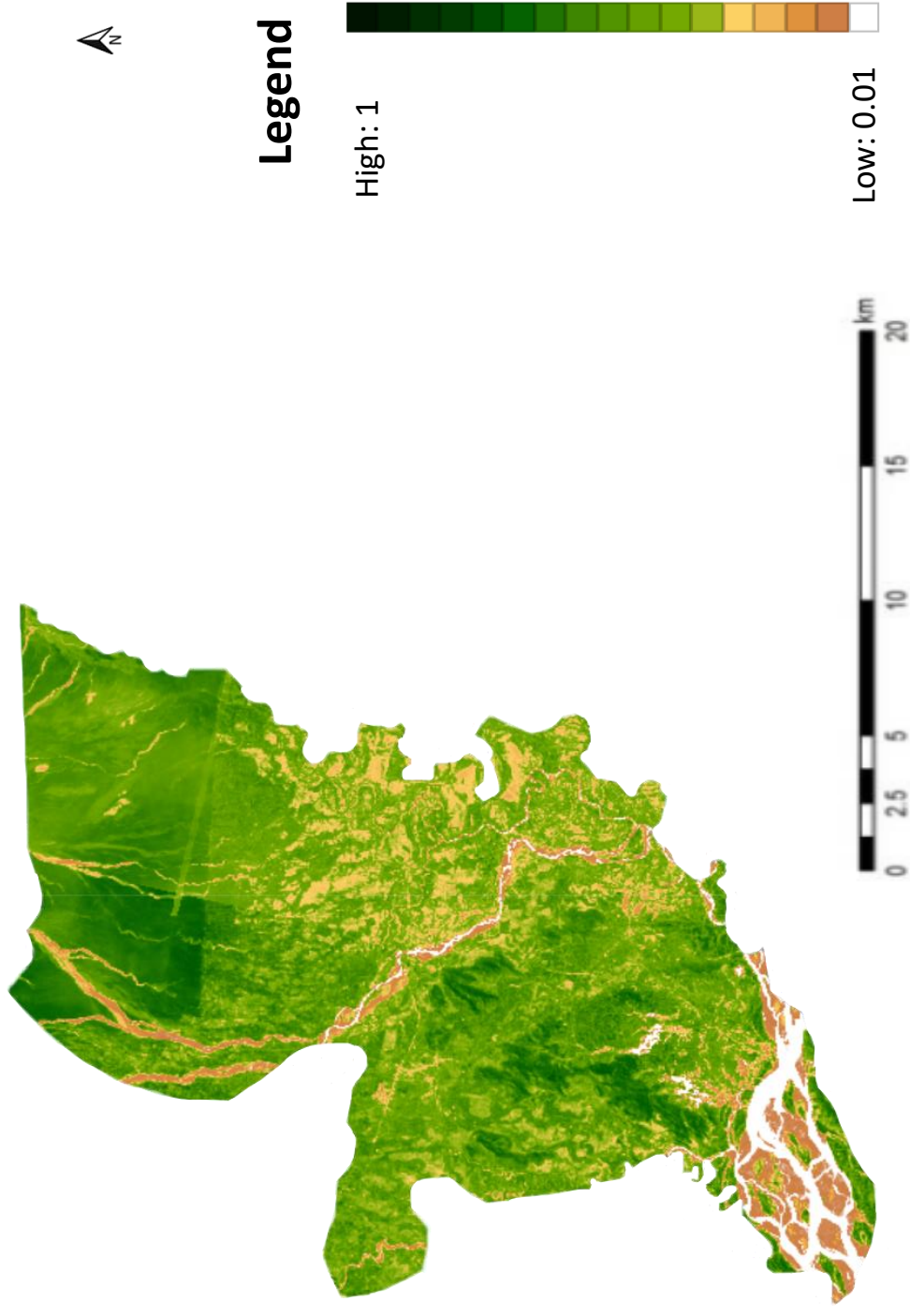
MAP SHOWING NORMALISED DIFFERENCE VEGETATION INDEX OF BONGAIGAON, 2003



MAP SHOWING NORMALISED DIFFERENCE VEGETATION INDEX OF BONGAIGAON, 2013

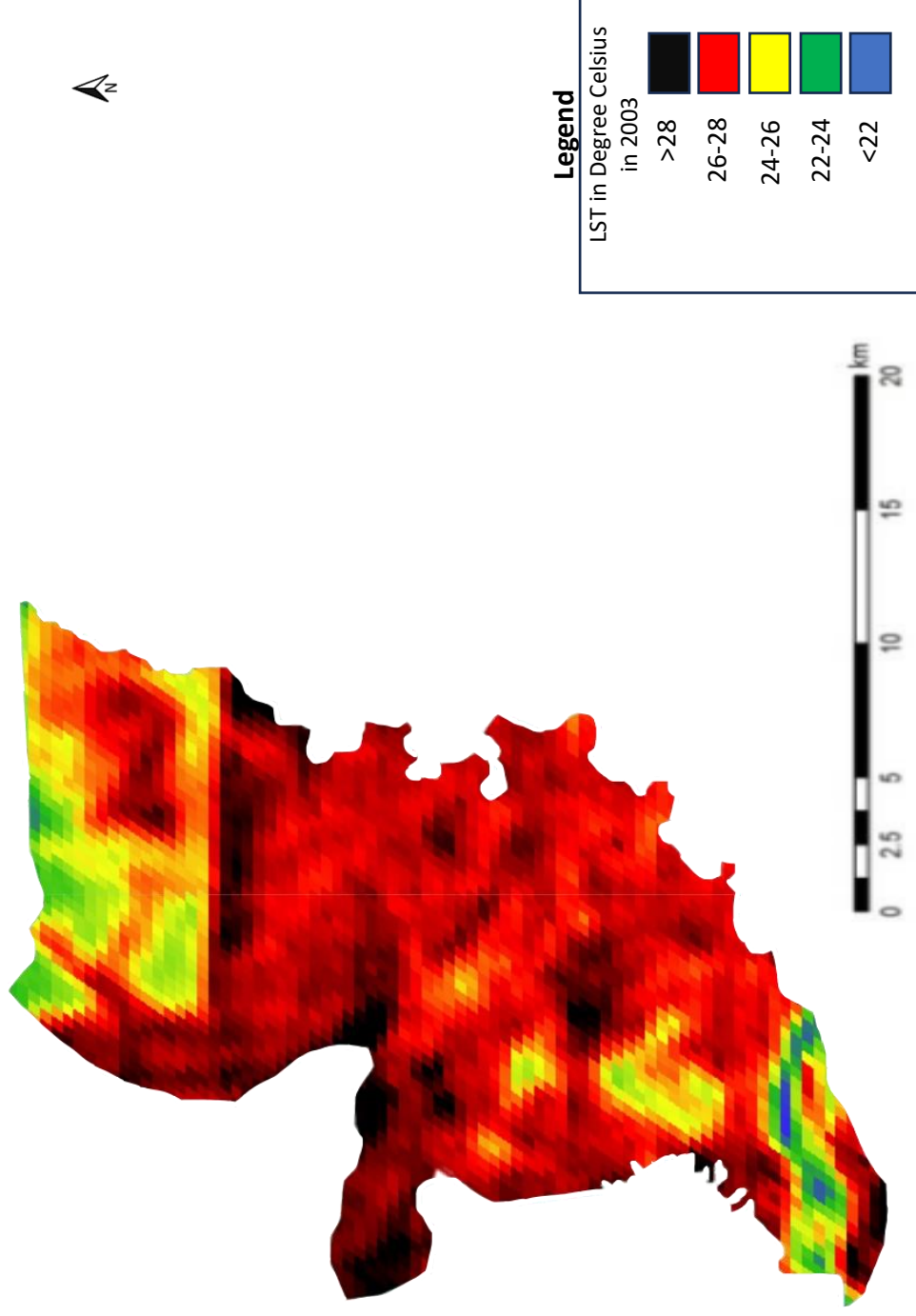


MAP SHOWING NORMALISED DIFFERENCE VEGETATION INDEX OF BONGAIGAON, 2023

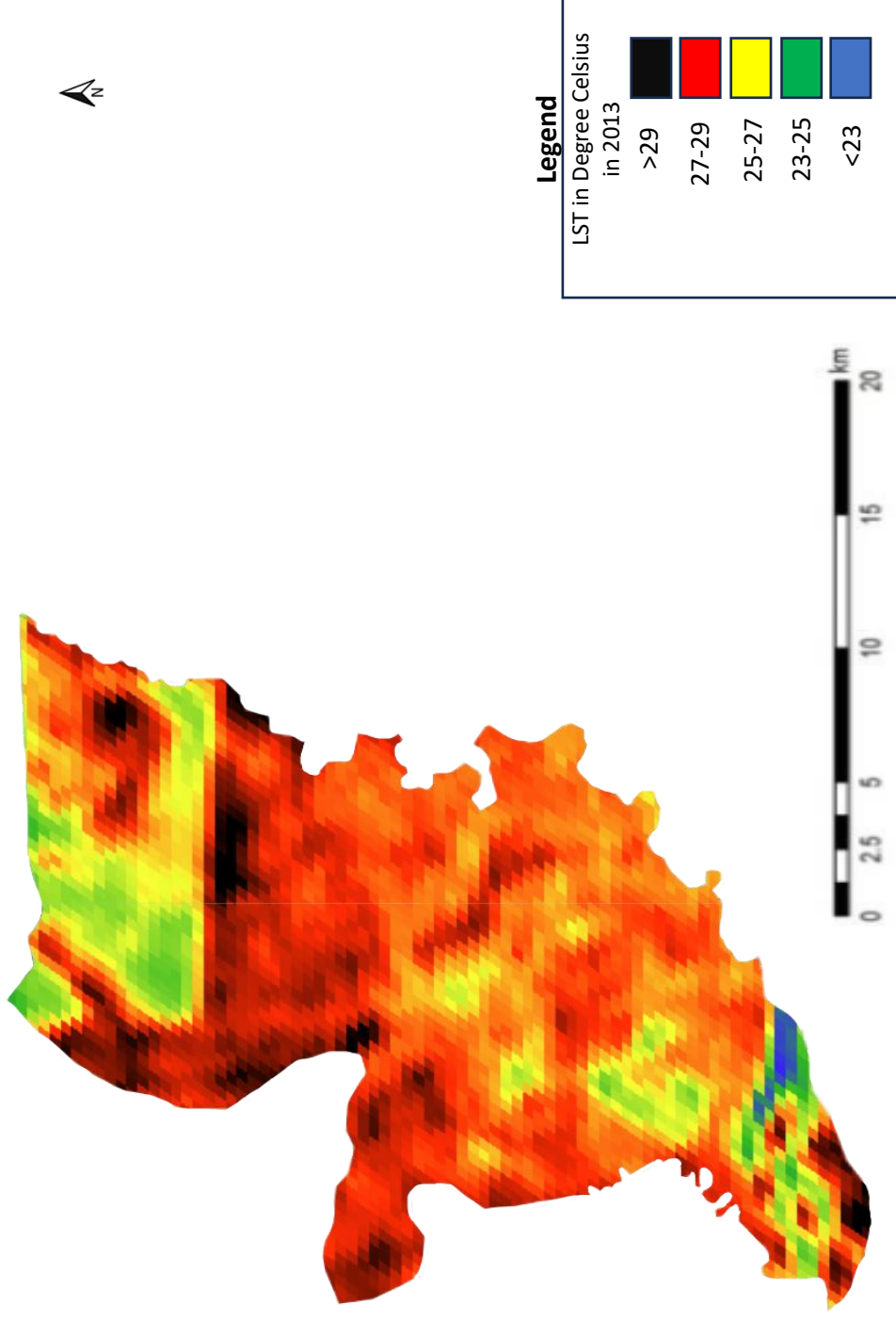


4.2 Land Surface Temperature (LST) Analysis:

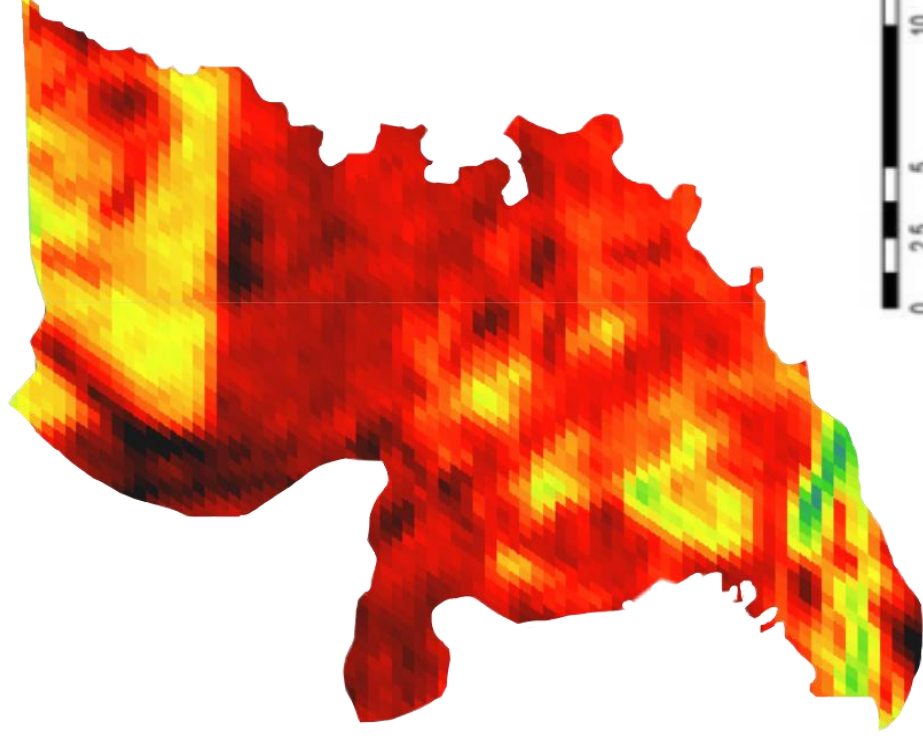
MAP SHOWING THE LAND SURFACE TEMPERATURE OF BONGAIGAON, 2003



MAP SHOWING THE LAND SURFACE TEMPERATURE OF BONGAIGAON, 2013



MAP SHOWING THE LAND SURFACE TEMPERATURE OF BONGAIGAON, 2023

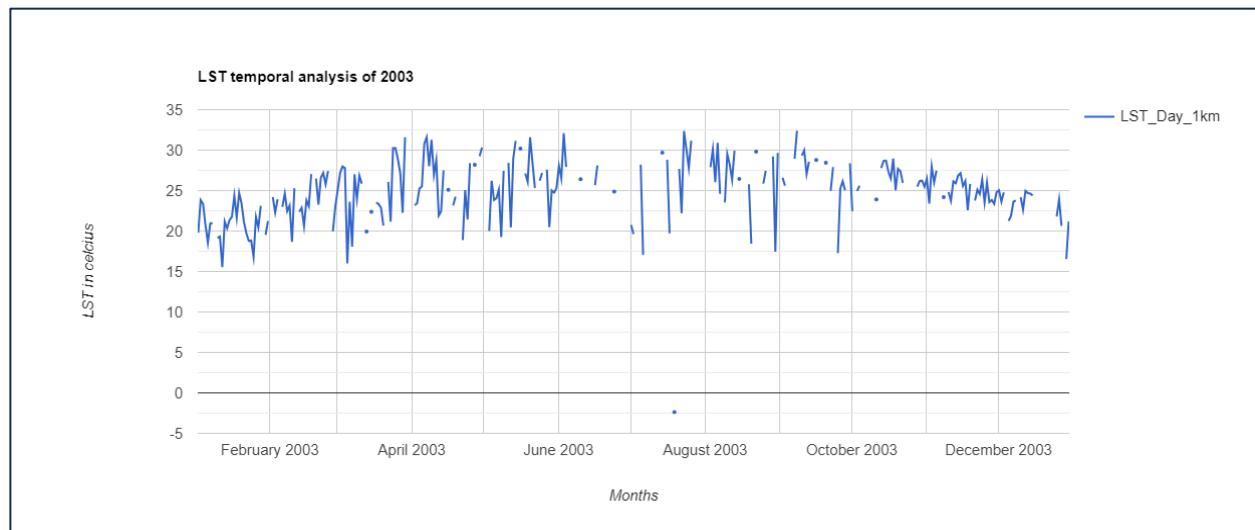


Legend

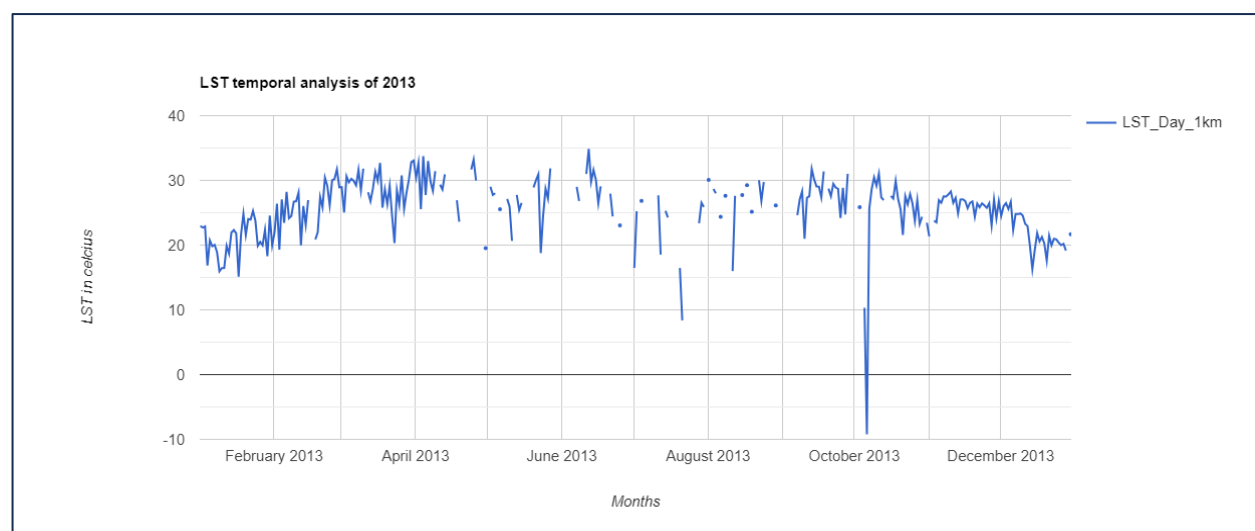
LST in Degree Celsius
in 2023



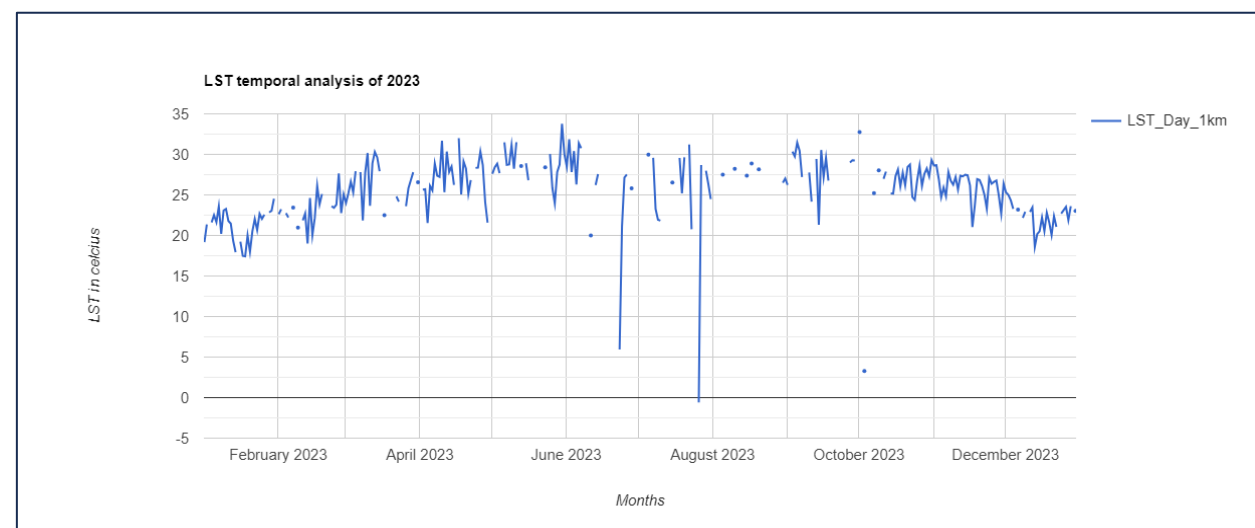
GRAPHICAL REPRESENTATION OF THE AVERAGE LAND SURFACE TEMPERATURE OF 2003, 2013 AND 2023



(a)



(b)



(c)

4.3. Discussion:

The analysis of land surface temperature (LST) data has highlighted a persistent and substantial increase in temperatures over recent decades, with a particularly notable surge observed in the mean temperatures of 2013 and 2023, which have risen to as high as 35 degrees celsius as we can see on the graph. This upward trend in temperature is closely linked to the expansion of urban areas, which exacerbates the urban heat island (UHI) effect. The LST data from 2003, 2013, and 2023 reveal the formation of increasingly hotter zones, predominantly concentrated in the central to southern parts of the region, where urban development is most pronounced. In contrast, the northernmost areas of the region, characterized by their high altitude and dense forest cover, exhibit significantly lower temperatures.

The analysis highlights the significant impact of urbanization on local climate conditions. The proliferation of built-up areas directly contributes to elevated land surface temperatures. The emergence of these heat zones is particularly worrisome due to their potential effects on the region's socio-economic and environmental well-being. Elevated temperatures can intensify heat stress, degrade air quality, and strain water resources, posing substantial risks to public health and the natural ecosystem. The analysis of land surface temperature (LST) serves as a crucial indicator of the region's environmental health, emphasizing the urgent need for measures to address the challenges posed by rising temperatures and urban expansion.

This study aware us for balancing urban growth with environmental preservation, highlighting the urgent need for sustainable strategies to manage urban heat islands and mitigate the impacts of climate change for the future of the Bongaigaon.

Chapter-5: Conclusion

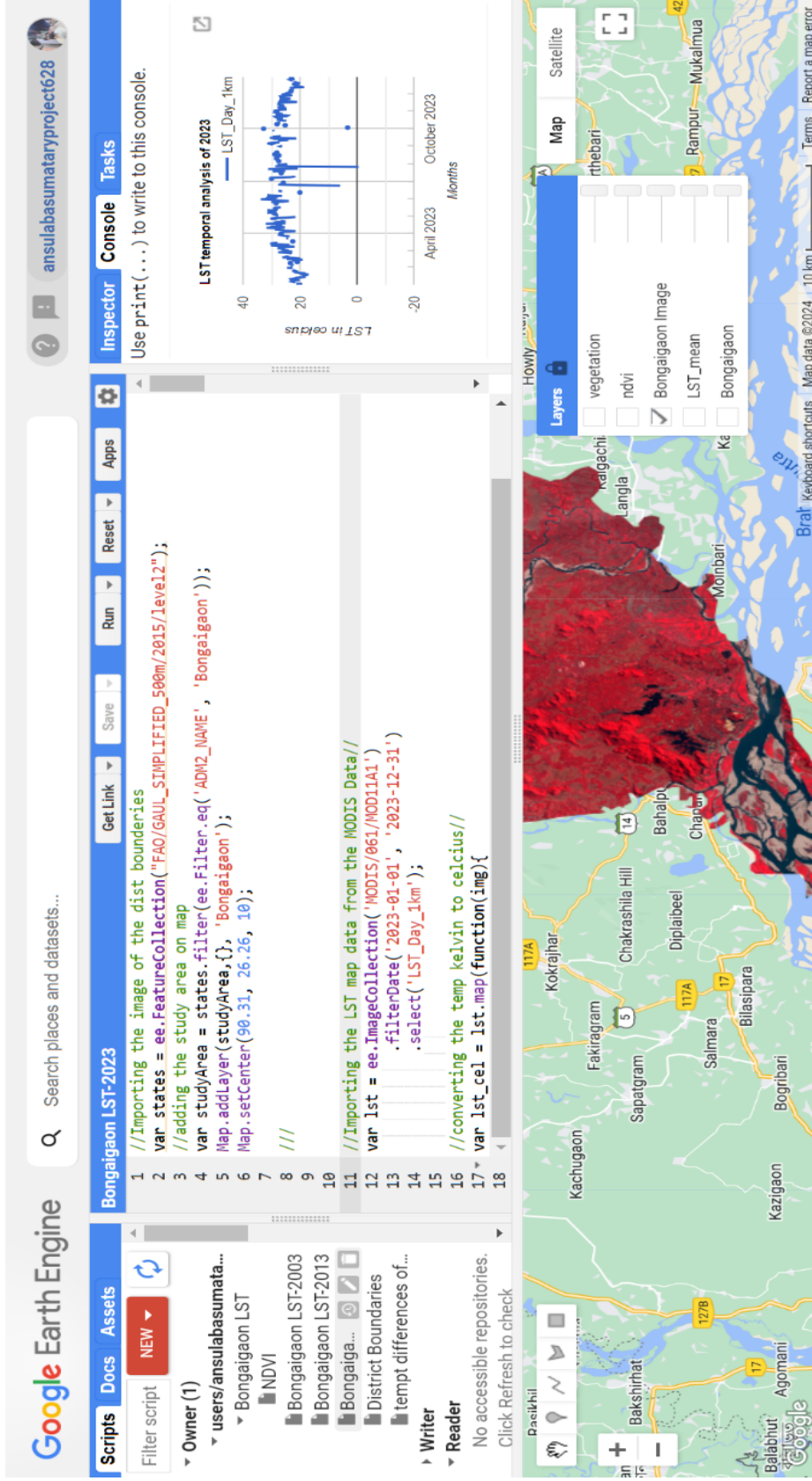
In this study, we deeply investigated the relationship between the Normalised Differences Vegetation Index (NDVI) and land surface temperature (LST) in the Bongaigaon District of Northeast India. Our research spanned the years 2003, 2013, and 2023. The resulting conclusions are multifaceted and extensive, intricately linked to the study's limitations, and they offer potential avenues for future research. Employing a supervised machine learning approach, our methodology yielded robust and accurate results. Notably, the NDVI mapping revealed significant shifts in forested areas, allowing us to analyze the impact of urban development and decreasing green cover. Simultaneously, the rising LST underscored a warming landscape influenced by urbanization.

We must recognize that the model's precision heavily relies on the quality and representativeness of input data, suggesting an avenue for improvement in future research. Projections reveal elevated land surface temperatures (LST), particularly in regions earmarked for urban expansion. This connection between urban growth and rising LST gains support from the interplay among LST, the Normalized Difference Built-up Index (NDBI), and the Normalized Difference Vegetation Index (NDVI). Notably, the study highlights that the influence of built-up areas on temperature fluctuations surpasses that of vegetation cover, emphasizing the pivotal role of urbanization in temperature escalation.

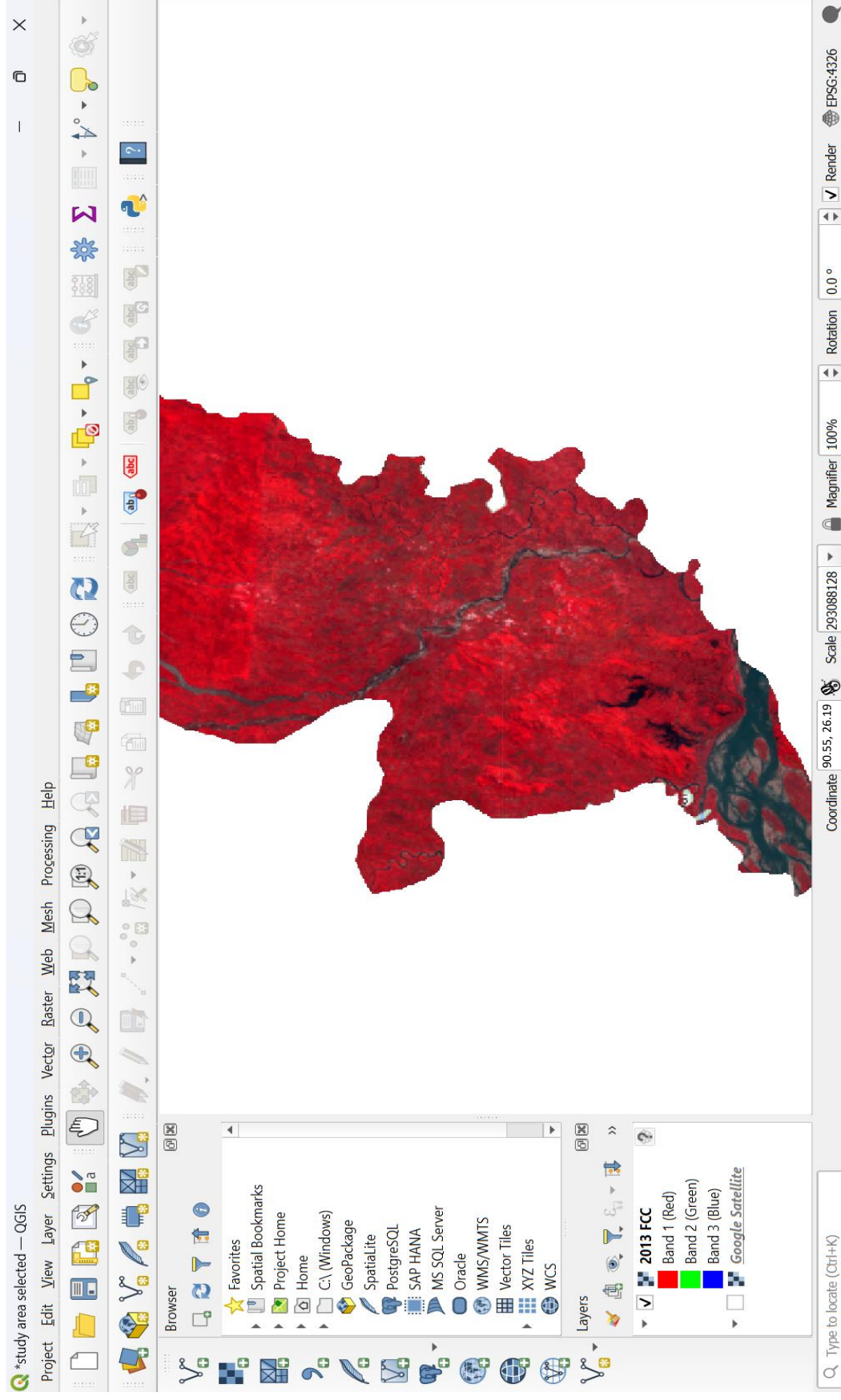
While the study offers valuable insights into urbanization, temperature trends, and urban heat island effects, it also underscores limitations and gaps. Relying on remote sensing data, despite its potency, introduces uncertainties tied to data quality and resolution. Furthermore, the unsystematic nature of the machine learning model introduces inherent uncertainty in its predictions. These limitations emphasize the importance of interpreting results cautiously and highlight the directions for future research.

Additionally, the study raises unresolved queries. While it underscores vegetation's significance in mitigating the urban heat island (UHI) effect, it does not delve into the specific contributions of various vegetation types and green infrastructure. Furthermore, exploring how urban planning strategies can optimally harness these natural resources to counteract the UHI effect demands further exploration.

In summary, this study offers a holistic grasp of the evolving Land Use and Land Cover (LSLU) and Land Surface Temperature (LST) dynamics in Bongaigaon. It underscores the significance of sustainable land management and urban planning in mitigating urban expansion's negative impacts. Simultaneously, it identifies unexplored avenues, steering the region toward a resilient and sustainable future.



(a) Selecting Study area, calculating NDVI, LST in Google Earth Engine



(b) Georeferencing the Study area, printing new layer etc

Reference:

1. : A. J. Prata, V. Caselles, C. Coll, J. A. Sobrino & C. Ottlé (1995) "Thermal remote sensing of land surface temperature from satellites: Current status and future prospects"
2. Kamal Kumar Talukdar Project Scientist, Assam Remote Sensing Application Centre, ASTE Council, Guwahati, Assam, India, (2020) "Land Surface Temperature Retrieval of Guwahati City and Suburbs, Assam, India using Landsat Data"
3. Upasana Choudhury, Suraj Kumar Singh, Anand Kumar, Gowhar Meraj, Pankaj Kumar and Shruti Kanga (2023), "Assessing Land Use/Land Cover Changes and Urban Heat Island Intensification: A Case Study of Kamrup Metropolitan District, Northeast India (2000–2032)"
4. Markus Neteler (2010) in his journal of LST "Estimating Daily Land Surface Temperatures in Mountainous Environments by Reconstructed MODIS LST DATA".
5. A.Benali, A.C. Carvalho, J.P. Nunes, N. Carvalhais and A. Santos (2012) in their journal of Air Surface Temperature "Estimating air surface temperature in Portugal using MODIS LST DATA"